

Chapter 10: Numerical Methods

Overview

Numerical linear algebra asks how matrix algorithms behave under finite precision and finite iteration. This chapter develops floating-point error, partial pivoting, conditioning, Jacobi and Gauss–Seidel iterations, the power method, and applications to regression, probability, and population models.

Original contexts

Examples use original numbers and fictional contexts involving hiking, driving, travel, music, badminton, and Spurs-inspired values. The mathematics remains the primary focus.

1 Floating-Point Arithmetic and Partial Pivoting

Normalized floating-point form

A base-ten normalized number is $\pm M10^k$ with $0.1 \leq M < 1$. A t -digit system stores only t significant digits of M .

Truncation discards all later digits. Rounding chooses the nearest stored number. Under round-half-to-even, an exact tie is resolved so that the final retained digit is even.

Stored values with four significant digits

exact number	normalized form	stored value
18.245	0.18245×10^2	18.24
18.255	0.18255×10^2	18.26
0.0099945	0.99945×10^{-2}	0.009994
0.0099955	0.99955×10^{-2}	0.009996

In the first and third rows the retained last digit is even, so an exact tie does not increase it. In the second and fourth rows the retained last digit is odd, so it increases.

If the exact result is x and the stored value is $\text{fl}(x)$, each arithmetic operation introduces a local perturbation. Repeated operations can accumulate error, but the principal danger is often amplification rather than simple accumulation.

Absolute and relative error

For exact x and approximation $\hat{x}$,

$$E_{\text{abs}} = |x - \hat{x}|, \quad E_{\text{rel}} = \frac{|x - \hat{x}|}{|x|} \quad (x \neq 0).$$

The floating-point model is

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u.$$

Subtraction of nearly equal numbers can cause catastrophic cancellation because most leading digits disappear.

A poor pivot amplifies rounding

Consider

$$0.002x_1 + 2x_2 - 2x_3 = 6.002,$$

$$3x_1 - 4x_2 - 4x_3 = -1,$$

$$x_1 + 4x_2 - 4x_3 = 13.$$

The exact solution is $(1, 2, -1)^T$. With four-significant-digit rounding and no pivoting,

$$m_{21} = 1500, \quad m_{31} = 500.$$

The rounded triangular matrix becomes approximately

$$\left[\begin{array}{ccc|c} 0.002 & 2 & -2 & 6.002 \\ 0 & -3004 & 2996 & -9004 \\ 0 & 0.1 & 2.5 & -2 \end{array} \right],$$

and back-substitution gives $(2.000, 2.199, -0.800)^T$.

The complete rounded elimination is

$$R_2 \leftarrow R_2 - 1500R_1,$$

$$R_3 \leftarrow R_3 - 500R_1,$$

which gives

$$\left[\begin{array}{ccc|c} 0.002 & 2 & -2 & 6.002 \\ 0 & -3004 & 2996 & -9004 \\ 0 & -996 & 996 & -2988 \end{array} \right].$$

The second multiplier is

$$m_{32} = \text{fl} \left(\frac{-996}{-3004} \right) = 0.3316.$$

After $R_3 \leftarrow R_3 - 0.3316R_2$, cancellation produces the small entries 0.1, 2.5, and -2 . These are differences of quantities near 996 and 2988, so several leading digits have been lost. The incorrect solution is a direct consequence of the poor first pivot.

Partial pivoting

At stage k , choose the active-column entry of largest magnitude and swap it into position (k, k) . Then every elimination multiplier satisfies

$$|m_{ik}| = \left| \frac{a_{ik}}{a_{kk}} \right| \leq 1.$$

For the same system, partial pivoting produces approximately

$$\left[\begin{array}{ccc|c} 3 & -4 & -4 & -1 \\ 0.0001 & 5.333 & -2.667 & 13.33 \\ 0 & 0 & -0.995 & 0.996 \end{array} \right]$$

and $(0.9973, 1.999, -1.001)^T$.

The pivoted passes are:

$$R_1 \leftrightarrow R_2, \quad m_{21} = 0.0006667, \quad m_{31} = 0.3333.$$

This gives

$$\left[\begin{array}{ccc|c} 3 & -4 & -4 & -1 \\ 0 & 2.003 & -1.997 & 6.003 \\ 0.0001 & 5.333 & -2.667 & 13.33 \end{array} \right].$$

In the second active column, 5.333 is larger than 2.003, so swap the last two rows and use

$$m_{32} = \text{fl} \left(\frac{2.003}{5.333} \right) = 0.3756.$$

The resulting third row is $[0, 0, -0.995 \mid 0.996]$. The multipliers remain below one in magnitude, and the final solution error is only about 2.7×10^{-3} in the infinity norm.

Residual and conditioning

For computed $\hat{x}$,

$$r = b - A\hat{x}.$$

A small residual is not sufficient for a small forward error when A is ill-conditioned. If $\hat{x}$ has residual $r = b - A\hat{x}$, then the exact error satisfies

$$A(x - \hat{x}) = r, \quad x - \hat{x} = A^{-1}r.$$

Thus

$$\|x - \hat{x}\| \leq \|A^{-1}\| \|r\|.$$

A small residual controls the forward error only when $\|A^{-1}\|$ is moderate. Define

$$\kappa(A) = \|A\| \|A^{-1}\|.$$

Right-hand-side perturbation bound

If $Ax = b$ and $A(x + \Delta x) = b + \Delta b$, then

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\Delta b\|}{\|b\|}.$$

Proof

Subtract the systems: $A\Delta x = \Delta b$, hence $\Delta x = A^{-1}\Delta b$ and $\|\Delta x\| \leq \|A^{-1}\| \|\Delta b\|$. Also $\|b\| = \|Ax\| \leq \|A\| \|x\|$, so $1/\|x\| \leq \|A\|/\|b\|$. Multiply.

An ill-conditioned two-equation system

Consider

$$\begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 2.0001 \end{bmatrix}.$$

The exact solution is $(1, 1)^T$. The two rows are nearly parallel, and the 2-norm condition number is about 4.0×10^4 . If the second right-hand side changes from 2.0001 to 2.0002, then subtraction of the equations gives $0.0001y = 0.0002$, so the new solution is $(0, 2)^T$. A relative data change of about 5×10^{-5} causes an order-one solution change.

10.1

Conditioning describes sensitivity of the problem. Stability describes how well an algorithm controls arithmetic error. Partial pivoting improves stability but cannot change the condition number.

2 Jacobi and Gauss–Seidel Iterations

Write $A = D + L + U$. Jacobi and Gauss–Seidel are

$$\begin{aligned}\mathbf{x}^{(k+1)} &= -D^{-1}(L + U)\mathbf{x}^{(k)} + D^{-1}\mathbf{b}, \\ \mathbf{x}^{(k+1)} &= -(D + L)^{-1}U\mathbf{x}^{(k)} + (D + L)^{-1}\mathbf{b}.\end{aligned}$$

If B is the iteration matrix and $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$, then

$$\mathbf{e}^{(k+1)} = B\mathbf{e}^{(k)}, \quad \mathbf{e}^{(k)} = B^k\mathbf{e}^{(0)}.$$

Thus convergence for every initial vector occurs exactly when $\rho(B) < 1$.

Spectral-radius convergence criterion

For a square iteration matrix B , $B^k \rightarrow 0$ if and only if $\rho(B) < 1$.

Detailed proof using Jordan form

Write $B = PJP^{-1}$, where J is block diagonal with Jordan blocks

$$J_m(\lambda) = \lambda I + N, \quad N^m = 0.$$

By the binomial theorem,

$$J_m(\lambda)^k = \sum_{r=0}^{m-1} \binom{k}{r} \lambda^{k-r} N^r.$$

If $|\lambda| < 1$, every term tends to zero because exponential decay dominates the polynomial factor $\binom{k}{r}$. Thus $J^k \rightarrow 0$ and $B^k = PJ^kP^{-1} \rightarrow 0$. Conversely, if $|\lambda| \geq 1$ for an eigenvalue with eigenvector v , then $B^k v = \lambda^k v$ does not tend to zero. Therefore $B^k \rightarrow 0$ exactly when every eigenvalue has magnitude below one.

Detailed system

$$\begin{aligned}10x_1 - x_2 + 2x_3 &= 6, \\ -x_1 + 11x_2 - x_3 &= 22, \\ 2x_1 - x_2 + 10x_3 &= -10.\end{aligned}$$

Updates are

$$x_1 = \frac{6 + x_2 - 2x_3}{10}, \quad x_2 = \frac{22 + x_1 + x_3}{11}, \quad x_3 = \frac{-10 - 2x_1 + x_2}{10}.$$

From zero, the first Jacobi iterates are

$$(0.6, 2, -1), \quad (1, 1.963636, -0.92), \quad (0.980364, 2.007273, -1.003636).$$

The Jacobi iteration table is

k	$x_1^{(k)}$	$x_2^{(k)}$	$x_3^{(k)}$
0	0	0	0
1	0.600000	2.000000	-1.000000
2	1.000000	1.963636	-0.920000
3	0.980364	2.007273	-1.003636
4	1.001455	1.997884	-0.995345

The first Gauss–Seidel iterates are

$$(0.6, 2.054545, -0.914545), \quad (0.988364, 2.006711, -0.997002), \\ (1.000071, 2.000279, -0.999986).$$

Strict diagonal dominance

Assume

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, \quad q = \max_i \frac{\sum_{j \neq i} |a_{ij}|}{|a_{ii}|} < 1.$$

For Jacobi,

$$|e_i^{(k+1)}| \leq q \|e^{(k)}\|_\infty,$$

so $\|e^{(k)}\|_\infty \leq q^k \|e^{(0)}\|_\infty \rightarrow 0$.

Gauss–Seidel convergence under strict diagonal dominance

The component error equation is

$$e_i^{(k+1)} = -\frac{1}{a_{ii}} \left(\sum_{j < i} a_{ij} e_j^{(k+1)} + \sum_{j > i} a_{ij} e_j^{(k)} \right).$$

Induct on i . Earlier updated errors are at most $q \|e^{(k)}\|_\infty$. Hence

$$|e_i^{(k+1)}| \leq \frac{q \sum_{j < i} |a_{ij}| + \sum_{j > i} |a_{ij}|}{|a_{ii}|} \|e^{(k)}\|_\infty \leq q \|e^{(k)}\|_\infty.$$

Taking the maximum proves contraction.

Divergence and row reordering

The displayed system

$$x + 4y = 5, \quad 3x + y = 4$$

has Jacobi iteration matrix $B = \begin{bmatrix} 0 & -4 \\ -3 & 0 \end{bmatrix}$ with eigenvalues $\pm\sqrt{12}$, so $\rho(B) > 1$ and the iteration diverges. Reordering the equations gives

$$3x + y = 4, \quad x + 4y = 5,$$

with iteration matrix $\tilde{B} = \begin{bmatrix} 0 & -1/3 \\ -1/4 & 0 \end{bmatrix}$ and $\rho(\tilde{B}) = 1/\sqrt{12} < 1$. The equations are unchanged; only the iterative splitting has changed.

A practical stopping rule checks both

$$\|x^{(k+1)} - x^{(k)}\|_\infty$$

and the residual $\|b - Ax^{(k+1)}\|_\infty$.

If a contraction estimate $\|e^{(k+1)}\| \leq q\|e^{(k)}\|$ is known, an a posteriori bound follows from

$$x^* - x^{(k)} = \sum_{j=k}^{\infty} (x^{(j+1)} - x^{(j)}).$$

The geometric-series estimate gives

$$\|x^* - x^{(k)}\| \leq \frac{q}{1-q} \|x^{(k)} - x^{(k-1)}\|.$$

This turns an observed iterate difference into an error bound.

10.2

Iterative methods are matrix fixed-point processes. Convergence is controlled by the iteration matrix, not merely by whether the original system has a unique solution.

3 The Power Method

Let A be diagonalizable with $|\lambda_1| > |\lambda_2| \geq \dots$. Expand

$$x^{(0)} = c_1 v_1 + \dots + c_n v_n, \quad c_1 \neq 0.$$

Then

$$A^k x^{(0)} = \lambda_1^k \left[c_1 v_1 + \sum_{j=2}^n c_j \left(\frac{\lambda_j}{\lambda_1} \right)^k v_j \right].$$

After scaling, the direction approaches v_1 . The dominant error decays like $|\lambda_2/\lambda_1|^k$.

Scaled algorithm and Rayleigh quotient

$$y^{(k+1)} = Ax^{(k)}, \quad \alpha_{k+1} = \|y^{(k+1)}\|_\infty, \quad x^{(k+1)} = y^{(k+1)} / \alpha_{k+1}.$$

For nonzero x ,

$$R_A(x) = \frac{x^T A x}{x^T x}.$$

If $Ax = \lambda x$, then $R_A(x) = \lambda$.

Why the Rayleigh quotient is exact on an eigenvector

For nonzero x satisfying $Ax = \lambda x$,

$$R_A(x) = \frac{x^T A x}{x^T x} = \frac{x^T (\lambda x)}{x^T x} = \lambda \frac{x^T x}{x^T x} = \lambda.$$

For an approximate eigenvector, the residual $Ax - R_A(x)x$ measures the failure of the eigen-equation.

Power iteration

For

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \quad x^{(0)} = (1, 0)^T,$$

the eigenvalues are 5 and 2. Scaled iterates begin

$$(1, 0.5)^T, (1, 0.777778)^T, (1, 0.906977)^T, (1, 0.962085)^T,$$

and approach $(1, 1)^T$. A longer table is

k	$x_1^{(k)}$	$x_2^{(k)}$	$R_A(x^{(k)})$
1	1	0.500000	5.000000
2	1	0.777778	5.076923
3	1	0.906977	5.041543
4	1	0.962085	5.018197
5	1	0.984718	5.007521
6	1	0.993869	5.003047

The scale factors also approach 5. Rayleigh quotients approach 5.

Two instructive failure cases

For $A = \text{diag}(5, 2)$ and $x^{(0)} = (0, 1)^T$, the dominant component is zero. Every iterate remains $(0, 1)^T$, so the method finds $\lambda = 2$, not 5. For $A = \text{diag}(1, -1)$ and $x^{(0)} = (1, 1)^T$, the two eigenvalues tie in magnitude and the iterates alternate directions rather than converge.

Failure can occur when the initial vector has no dominant component or when dominant eigenvalues tie in magnitude. Convergence is slow when $|\lambda_2/\lambda_1|$ is near one. Check the eigen-residual

$$\|Ax - R_A(x)x\|.$$

10.3

The power method converts repeated matrix multiplication into an eigenvector filter. Scaling controls magnitude; the Rayleigh quotient estimates the eigenvalue.

4 Applications

Quadratic regression

For $p(x) = a_0 + a_1x + a_2x^2$, minimizing squared residuals yields

$$\begin{bmatrix} n & \sum x & \sum x^2 \\ \sum x & \sum x^2 & \sum x^3 \\ \sum x^2 & \sum x^3 & \sum x^4 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} \sum y \\ \sum xy \\ \sum x^2y \end{bmatrix}.$$

For fictional hiking data $(0, 120), (1, 182), (2, 255), (3, 329), (4, 388), (5, 415)$, the system is

$$\begin{bmatrix} 6 & 15 & 55 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1689 \\ 5306 \\ 20746 \end{bmatrix}.$$

Partial-pivoting elimination begins by swapping the largest first-column entry, 55, into the first row:

$$\left[\begin{array}{ccc|c} 55 & 225 & 979 & 20746 \\ 15 & 55 & 225 & 5306 \\ 6 & 15 & 55 & 1689 \end{array} \right].$$

The first multipliers are $15/55$ and $6/55$. After the first pass and a second active-row pivot, back-substitution gives

$$a_2 = -4.125, \quad a_1 = 82.539286, \quad a_0 = 112.964286.$$

Therefore

$$p(x) = 112.964286 + 82.539286x - 4.125x^2,$$

and $p(6) \approx 459.7$. The residual vector is orthogonal to the three columns 1, x , and x^2 , which is another form of the normal equations.

Gauss–Seidel probability model

Let p_i be the probability of reaching a scenic overlook before a failure exit. Suppose

$$p_1 = \frac{1}{2}p_2, \quad p_2 = \frac{1}{3}(p_1 + p_3 + p_4), \quad p_3 = \frac{1}{2}(p_2 + 1), \quad p_4 = \frac{1}{3}(p_2 + 1).$$

Writing the equations in matrix form gives

$$\begin{bmatrix} 1 & -1/2 & 0 & 0 \\ -1/3 & 1 & -1/3 & -1/3 \\ 0 & -1/2 & 1 & 0 \\ 0 & -1/3 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1/2 \\ 1/3 \end{bmatrix}.$$

The exact solution is

$$(p_1, p_2, p_3, p_4) = \left(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{1}{2}\right).$$

Starting Gauss–Seidel at zero produces

k	p_1	p_2	p_3	p_4
1	0	0	0.500000	0.333333
2	0	0.277778	0.638889	0.425926
3	0.138889	0.401235	0.700617	0.467078
4	0.200617	0.456104	0.728052	0.485368
6	0.240245	0.491329	0.745665	0.497110

and converges to the exact probabilities. The equations express the law of total probability: an interior probability is the average of the probabilities after the next random move.

Leslie population model

Use

$$L = \begin{bmatrix} 0 & 1.1 & 1.6 & 0.5 \\ 0.72 & 0 & 0 & 0 \\ 0 & 0.65 & 0 & 0 \\ 0 & 0 & 0.55 & 0 \end{bmatrix}.$$

If $n_{k+1} = Ln_k$, a stable distribution satisfies $Lv = \lambda v$. Starting from equal age counts and normalizing the total after each multiplication produces

k	class 1	class 2	class 3	class 4
1	0.6154	0.1846	0.1667	0.0333
2	0.4695	0.3418	0.1026	0.0861
3	0.4898	0.2747	0.1832	0.0523
5	0.4869	0.2911	0.1482	0.0738

The proportions settle as the dominant eigendirection emerges. Power iteration gives dominant growth factor about 1.221244 and normalized proportions

$$(0.48908, 0.28834, 0.15347, 0.06912)^T.$$

Interpretive caution

Algorithmic convergence does not validate the scientific model. Regression extrapolation, fixed transition probabilities, and time-invariant fertility or survival are separate assumptions that require domain evidence.

Chapter terminology

Term	Meaning
Floating-point number	Finite stored approximation in normalized scientific notation
Round-half-to-even	Unbiased tie-breaking convention
Partial pivoting	Largest active-column pivot selection
Residual	$b - A\hat{x}$
Condition number	Worst-case sensitivity factor
Jacobi method	Simultaneous component update iteration
Gauss–Seidel method	Sequential newest-value update iteration
Iteration matrix	Matrix governing error propagation
Spectral radius	Maximum eigenvalue magnitude
Dominant eigenvalue	Unique eigenvalue of largest magnitude
Rayleigh quotient	Eigenvalue estimate $x^T Ax / x^T x$
Leslie matrix	Age-structured population transition matrix